FOUNDATIONAL LEARNING CHALLENGE · SYSTEM CAPACITY

SAMVAAD

Turning every mentor visit into one evidence-backed, followed-up coaching cycle

An AI copilot for CRPs, BRPs, ARPs and NIPUN mentors — plan → observe → coach → demonstrate → follow up — built to run inside the existing NIPUN Bharat block/cluster structure.

Team: [Team name] · [Institution]

Submission: September 2026 · “SAMVAAD” (संवाद, dialogue) is a working name

Every claim in this deck is tagged:

FACT DERIVED ASSUMPTION HYPOTHESIS TARGET DESIGN

Numbered sources → source register (appendix)

PROBLEM UNDERSTANDING 01

India built a mentor cadre to fix foundational learning — but the system counts mentor visits, not the teaching those visits change

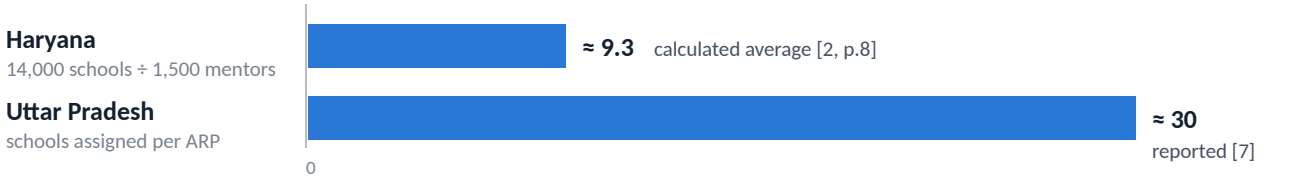
10,13,322
government schools, taught by **51.5 lakh** teachers (2024–25) [3] FACT

23.4%
of Std III children in rural govt schools can read a Std II text — **~3 in 4 cannot** [4] FACT DERIVED

What the system measures. Samagra Shiksha monitoring indicators for BRC/CRC support: "% of schools visited **5 to 6 times** in a year" and "% never visited" [5] FACT. In Haryana, mentor success is "measured by the number of **compliant visits**" [2, p.10] FACT.

> "The quality of FLN delivery … depends critically on the academic mentoring and coaching capacity of CRPs and BRPs. Yet evidence shows **monitoring systems have been reduced to data collection exercises**."
>
> Central Square Foundation, *NIPUN Bharat Mission at Five*, Jun 2026 [6] FACT

Same role, very different load — schools per mentor



Haryana value is DERIVED (actual allocation varies); UP value is reported assignment FACT. Different states and cadre definitions — shown to illustrate a ~3× spread in load, not a like-for-like comparison.

Inside a 2-hour mentor visit protocol (minutes)



Structured protocol recommended for NIPUN 2.0 [6]; UP's ARP model uses a "structured 2-hour school visit protocol" [7] FACT. At ~30 schools, that is ≥60 on-site hours/month before travel DERIVED.

Why existing mechanisms fall short

Mentor apps record the visit. A record is not a follow-up [2]

Spot-assessment data is "rarely used for … targeted teacher support" [6]

DIKSHA content is not tied to what the mentor saw in that classroom [15]

Coaching works, but effects shrink as programmes scale [9]

Problem: Mentors get a short, fixed window per visit, a caseload of roughly 9–30 schools, and reporting systems that reward coverage. Observation data gets collected, but it rarely turns into **a specific next step that is followed up** with the teacher. That missing link is the capacity gap we target.

Sources: [2] Reach Alliance/Oxford, *Teacher Mentorship in India*, Jan 2024, pp.8,10 · [3] MoE, UDISE+ 2024–25, Table 1 p.28 · [4] ASER Centre, ASER 2024 National Findings, p.47 · [5] Samagra Shiksha Programmatic & Financial Norms · [6] CSF, NIPUN Bharat Mission at Five, 23 Jun 2026 · [7] CSF, ARP Recruitment in UP, 22 Jan 2026 · [9] Kraft et al. 2018 · [15] PIB DIKSHA 2026

USERS & CONTEXT 02

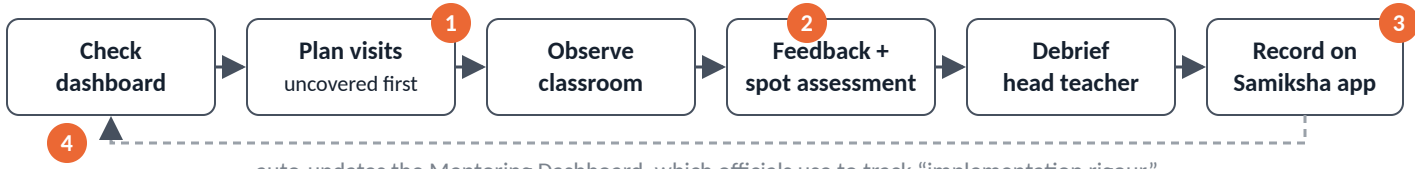
Mentors sit at the thinnest point of a 5-tier system, and one person plays three roles on every visit

NIPUN Bharat 5-tier structure — who owns mentoring [1] FACT

National	DoSE&L mission; NCERT/NRG guide ATFs on "mentor identification and their mapping" (p.16)
State	ATF: "identification of mentors, build their capacity and map them to the teachers/schools" (p.23); SCERT builds capacity of resource persons (p.22)
District	DIET: handhold & train mentors, meet them on challenges (p.30); DPMU: dashboards, MIS, KPIs (p.29)
Block / Cluster	**BEO**: "ensure regular and adequate on-site support to teachers by mentors/BRC/CRC" (p.35). **BRC/CRC/Mentors**: in-service training, **"onsite academic support ... in adopting new pedagogical practices"**, monitoring, community drives (p.36)
School	Head teacher & teachers implement curriculum, pedagogy, assessment (p.37)

The BRC essay provided with the brief adds that BRCs oversee CRCs, which "directly assist schools and teachers," and that understaffing is a recurring challenge [19].

A mentor's day today — Haryana case (state-specific, [2] pp.7–8)



1 · Coverage-driven planning. Visit plans prioritise "uncovered schools" [2,p.8] — not which teacher most needs which support. FACT

2 · Adviser vs auditor vs mediator. "Sometimes I mentor, audit, help, mediate ..." [2,p.11–12]. Bihar observation: 3 of 4 CRCCs gave no academic feedback; <21% of school time spent in classrooms (n=4) [8] FACT

3 · Reporting load. "A lot of time [must be] spent in reporting, adding photos, filling questionnaires" — making it hard to give 50% of time to mentoring [2,p.13] FACT

4 · Loop ends at the record. Success = "number of compliant visits" [2,p.10]. None of the KPIs we reviewed track whether the teacher acted on the feedback [2][5] DERIVED

What already works, and should be kept: teachers value repeat visits from the same mentor [2,p.13,17]; mentors are trained in "sandwich" and ask–describe–ask feedback [2,p.11]; dashboards and apps already exist. **The design brief is to strengthen this loop, not to replace it.** DESIGN

Three roles, one visit [2, p.9] FACT

Adviser
Guides pedagogy; teachers take up the advice — "Every time mentor visits, [I] learn something new" [2,p.11]

Auditor
Checks programme compliance, records and infrastructure — "Mentorship, not monitoring. We need to be a helper" [2,p.12]

Mediator
Two-way channel between teachers and block/district: relays issues up and priorities down [2,p.12]

Design implication: AI takes on the auditor's paperwork so the visit time goes to the adviser role. DESIGN

Sources: [1] MoE/DoSE&L, *NIPUN Bharat Mission: Stakeholders' Roles & Responsibilities* (printed pp.16–37) · [2] Reach Alliance/Oxford 2024 — Haryana, 20 interviews, Samagra-selected sample (p.9) · [5] Samagra Shiksha Norms · [8] Accountability Initiative, CRCCs in Bihar (4 CRCCs, 2 districts, 2014–16) · [19] A.C. Mehta, Intro to BRC 2023 (secondary)

USERS & CONTEXT 03

Five users share one leak: the step from what a mentor observes to what a teacher does next

USER	OFFICIAL JOB [1]	EVIDENCED PAIN	WHAT AI TAKES ON
Mentor / CRP / BRP / ARP	Onsite academic support, training, monitoring (p.36)	Admin and reporting crowd out mentoring [2,p.13]; 9–30 schools each [2] [7]; 20-min feedback slot [6]	Drafts feedback, demo script and visit report
Teacher	Implement FLN pedagogy & assessment (p.37)	Overburdened; wants advice that can be done, not "ideal solutions" [2,p.12]; many travel >50 km a day [2,p.17]	One concrete next step, in their language, as a voice note
Head teacher	Support academic innovation, pedagogy (p.37)	Included in the end-of-visit talk [2,p.8]; *HT experience not studied in [2]: evidence gap*	School-level list of agreed next steps
BEO	Ensure regular, adequate on-site support (p.35)	Tracks visit counts [5]; sees no view of which practices are improving	Block view of practice gaps (aggregate only)
DIET / DPMU	Train mentors; dashboards & KPIs (pp.29–30)	Visit data "rarely used for instructional decision-making or targeted teacher support" [6]	Training needs taken from observed gaps

We do not assume smartphones, steady internet, English or free time for any user. 36.5% of schools report no internet [3] DERIVED.

Which sub-problem to solve? Opportunity scoring (1–5)

	SEV	FRQ	USR	GAP	AI	FEA	MEA	DIF	DAT	/45
SP1 Visit & task prioritisation	3	4	4	3	4	4	4	2	4	32
SP2 Observation quality	4	5	4	4	3	3	4	4	2	33
SP3 Actionable feedback + follow-up	5	5	5	5	4	4	4	4	3	39
SP4 Practice demonstrations	3	3	4	3	3	4	2	3	3	28
SP5 Data use & reporting burden	4	5	5	4	4	3	4	3	3	35

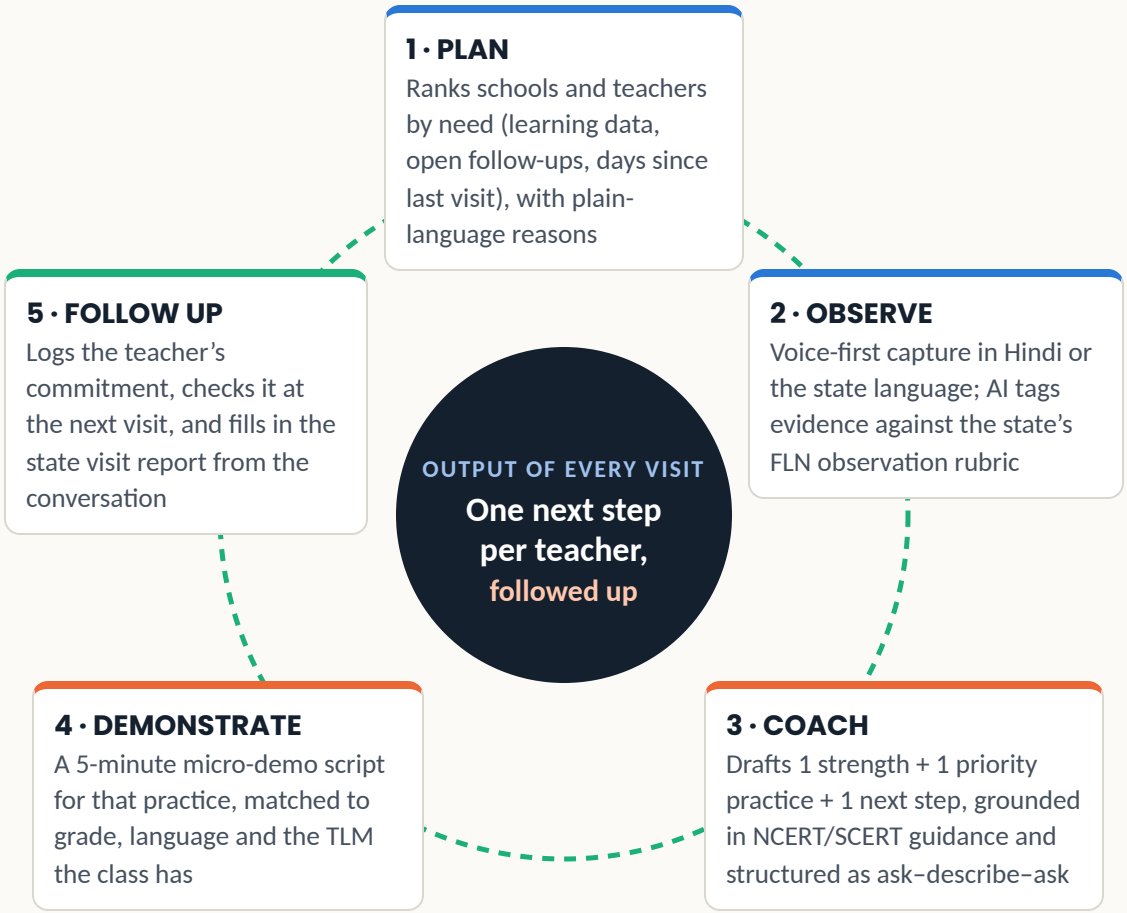
Team scoring, equal weights, max 45. The evidence behind each score is in Dossier Part B ASSUMPTION. Criteria: SEV severity · FRQ frequency · USR users affected · GAP current system gap · AI AI suitability · FEA feasibility · MEA measurability · DIF differentiation · DAT data availability.

Selected (SP3 + SP5): "Within a 2-hour visit, help a mentor turn classroom evidence into **one specific, practicable next step** for the teacher, deliver it well, show how it is done, and **follow it up at the next visit**, with the paperwork written automatically." Prioritisation (SP1) and demonstrations (SP4) come in only where they feed this loop. DESIGN

Sources: [1] NIPUN Bharat Roles & Responsibilities, pp.29–37 · [2] Reach Alliance/Oxford 2024 (Haryana) · [3] UDISE+ 2024–25 Table 1 (63.5% of schools with internet) · [5] Samagra Shiksha Norms · [6] CSF 2026 · [7] CSF 2026 (UP ARPs)

PROPOSED SOLUTION 04

SAMVAAD: an AI copilot that turns each visit into one evidence-backed coaching cycle — the AI drafts, the mentor decides



Division of labour DESIGN

AI DOES	MENTOR DOES
Transcribes, tags evidence, finds the matching practice card	Chooses the one practice to focus on
Drafts feedback and demo script in the local language	Edits, delivers face to face, models the practice
Pre-fills the report; schedules the follow-up	Reviews and submits; keeps the relationship
Aggregates practice gaps for the block and DIET	BEO/DIET plan training and support

Non-negotiables DESIGN

- **No AI teacher scores or rankings.** Outputs are evidence and suggestions.
- **No video, no face recognition.** Audio is opt-in, redacted and deleted on a schedule.
- **Works offline at school**; heavy AI runs after sync.
- **Plugs into existing state apps and dashboards** (e.g., Samiksha-type), not a parallel system.

Why this has to be AI: without speech-to-evidence, grounded generation and auto-reporting, the loop adds work for the mentor. The Haryana evidence shows paperwork is already the constraint [2,p.13].

Sources: [2] Reach Alliance/Oxford 2024, pp.11,13 (feedback models; reporting load). Module design is the team's proposal DESIGN; its effects are tested in the pilot (slide 9).

PROPOSED SOLUTION 05

One mentor, one school visit — before and after SAMVAAD

	① NIGHT BEFORE	② OBSERVE (40 MIN)	③ FEEDBACK (20 MIN)	④ DEMONSTRATE	⑤ AFTER VISIT	⑥ NEXT VISIT
Today's pain	Plan by coverage: "uncovered schools" [2]	Handwritten notes; mentor is also auditing [2][8]	Quality depends on the mentor; feedback often generic or absent [8]	Needs a ready, context-fit activity on the spot	Reports, photos, questionnaires [2,p.13]	No record of what was agreed last time
AI does	Ranked list with reasons; caches rubric and practice cards for offline use	Records opt-in audio; one-tap checklist; voice notes → evidence tags	Draft: 1 strength · 1 practice · 1 next step, with the source it came from	5-min demo script from the DIET-approved library	Pre-filled report; teacher gets a voice note in their language	Opens last commitment; checks for evidence of change
Mentor benefit	Arrives knowing why this school	Watches the class instead of writing	Specific, practicable advice	Shows the practice instead of only describing it	Less paperwork HYPOTHESIS	Continuity across visits
Data created	Priority score + reasons	Transcript, rubric tags + confidence	Feedback text, edit distance	Practice ID used	Report, commitment	Follow-up status

Illustrative feedback card — wireframe, not real teacher data

Grade 2 · Hindi · Visit 3 · Draft — mentor approval required

Strength: Children read the story aloud in pairs (00:12:40).

Priority practice: Ask questions about meaning after reading.

Next step: 3 "why/what next" questions after each read-aloud this week.

अगला कदम: हर कहानी के बाद 3 "क्यों/आगे क्या" प्रश्न पूछें ।

Source: state FLN teacher-guide card #R-14 · Follow-up due: next visit

Built on existing practice. Haryana mentors are already trained on sandwich and ask–describe–ask feedback [2,p.11], and UP ARPs already do demonstrations, spot assessments and feedback [7]. SAMVAAD structures these steps and does the paperwork. It does not introduce a new pedagogy. DESIGN

Realism check: audio capture is optional, and the checklist + voice-note path works without it. Female teachers may be uncomfortable in one-to-one sessions with male mentors [2,p.14], so a teacher can ask for the voice-note-only mode or for group feedback. DESIGN

Relationship first: the research shows trust builds over repeat visits by the same mentor [2,p.13]. The planner keeps teacher–mentor pairs stable, and every new mentor gets a school-context brief before the first visit, as the report recommends [2,p.17].

Sources: [2] Reach Alliance/Oxford 2024 (Haryana), pp.8,11,13,14,17 · [7] CSF 2026, UP ARP role (demonstrations, spot assessments, feedback) · [8] Accountability Initiative, Bihar CRCCs (n=4)

INNOVATION & CREATIVITY 06

The AI engine: five narrow models, each with a named human checkpoint — no single model "judges" teaching

INPUTS

Government data
UDISE+ school profile · state spot/FLN assessment results · visit history

Captured at visit
Opt-in classroom audio · mentor voice notes · checklist taps · photo of student work

Knowledge base
NCERT/SCERT FLN guides, state rubric, DIET-approved practice cards, DIKSHA items

PRE-PROCESSING

On device: voice-activity detection, Opus compression, redacting children's names
After sync: Indic ASR [16], OCR of worksheets

ENGINE	METHOD	OUTPUT	HUMAN CHECKPOINT	GUARDRAIL
E1 **Visit prioritiser**	Interpretable ranking: rules first, then gradient-boosted trees once outcome data exist	Ranked schools/teachers with the top-3 reasons	Mentor accepts, edits or overrides	No demographic features; keeps same-mentor continuity
E2 **Observation tagger**	ASR → segment classifier (fine-tuned small multilingual encoder) mapped to rubric items	Evidence snippets with timestamps and confidence	Mentor confirms tags	Low-confidence tags hidden; **never a teacher score**
E3 **Feedback composer**	LLM + retrieval (RAG) over approved guidance; fixed template (ask–describe–ask)	1 strength · 1 practice · 1 next step, with citation	Mentor edits and delivers it in person	Must cite a card; uncited text is blocked
E4 **Demo generator**	RAG over DIET-curated practice cards; adapts grade, language, TLM	5-min demo script + DIKSHA link	Mentor models it; DIET approves the card library	Generation limited to the library
E5 **Follow-up & report**	Structured extraction → state form schema; reminder scheduler	Pre-filled visit report; commitment tracker	Mentor reviews and submits	Audit log of every AI field

Learning loop

Mentor edits, accepted drafts and next-visit follow-up status become labelled data. These recalibrate E1 and improve prompts and cards through **offline, governed evaluation**. No live self-training on teacher data. DESIGN

Why narrow + grounded (evidence)

- Zero-shot ChatGPT coaching: **82%** of suggestions pointed to things the teacher was already doing [14] FACT → grounding and mentor edits are required.
- Automated feedback on teacher talk raised uptake of student ideas by **13%** in an RCT with 1,136 instructors (US, online) [13] FACT.
- In Kenya, coaching software suggests key points from learners' fluency at national scale [11] FACT.

Novelty: existing Indian tools put AI into *assessment* (e.g., UP's NIPUN Lakshya uses speech recognition for oral reading fluency [10]) or into *teacher lesson planning* (Shiksha Copilot [12]). SAMVAAD puts AI into the **mentor's coaching conversation and its follow-up**, which the reviewed sources describe as the weakest link [6].

Sources: [6] CSF 2026 · [10] Samagra, NIPUN Lakshya App, Jun 2024 · [11] Piper et al., IJEDICT 13(3), 2017 · [12] Microsoft, Shiksha Copilot, Feb 2024 · [13] Demszky et al., EEPA 2024 · [14] Wang & Demszky, BEA/ACL 2023 · [16] AI4Bharat IndicConformer (22 languages, MIT).
Model choices are DESIGN to be validated in Phase 0.

Existing tools either digitise monitoring or help teachers plan. None we found closes the mentor's observe-to-follow-up loop

CAPABILITY	DIKSHA [15]	HARYANA SAMIKSHA + DASHBOARD [2]	UP NIPUN LAKSHYA [10]	GUJARAT VSK [18]	SHIKSHA COPILOT [12]	TANGERINE:TUTOR KENYA [11]	WORLD BANK TEACH [17]	SAMVAAD (PROPOSED)
Need-based visit prioritisation	○	◖	○	○	○	○	○	◇
Classroom observation capture	○	●	○	◖	○	●	●	◇
AI-drafted teacher feedback	○	○	○	○	○	◖	○	◇
Demonstration support for mentor	◖	○	○	○	◖	○	○	◇
Follow-up tracking of feedback	○	◖	○	○	○	○	○	◇
Student data used in coaching	○	◖	●	●	○	●	○	◇
Indian-language support	●	○	◖	○	●	○	○	◇
Offline workflow	●	○	○	○	○	○	○	◇
Government-system integration	●	●	●	●	◖	●	◖	◇
AI actually used for	none cited	none cited	ORF speech	dropout pred. (pilot)	lesson plans	rule-based tips	none	**coaching loop (5 engines)**

Legend: ● documented in the cited source · ◖ partial / manual · ○ **not documented in the cited source** (not a claim that it is absent) · ◇ proposed, to be validated in pilot.

Where we differentiate: AI-drafted, grounded feedback + follow-up tracking + mentor-side demonstrations, in the mentor's language, offline at school. **Where we don't:** content (use DIKSHA), assessment AI (reuse state tools such as NIPUN Lakshya), dashboards (feed existing state MIS).

Sources: [2] Reach Alliance 2024 pp.7–8 · [10] Samagra Governance 2024 · [11] Piper et al. 2017 · [12] Microsoft 2024 · [15] PIB DIKSHA factsheet, 28 Jun 2026 (135 languages; offline download) · [17] World Bank, Teach Primary brief (2×15-min observations; 28 behaviours) · [18] NITI Frontier Tech, Gujarat VSK (2023). Ratings reflect cited sources only; full notes in Dossier Part H.

TECHNOLOGY & DATA FEASIBILITY 08

Built on data states already collect, and designed for a school with no signal

DATASET	OWNER / SOURCE	OBTAINABLE?	USED BY	PRIVACY
School profile	UDISE+ (MoE) [3]	Yes — national, school-level	E1	Low
FLN spot / state assessments	State apps (e.g., UP: 5 students per mentor visit [10])	State-dependent; needs data-sharing MoU ASSUMPTION	E1, E3 (class-level only)	Medium — child data aggregated
Visit history	State mentor apps (e.g., Samiksha [2])	State-dependent; API or CSV export	E1, E5	Medium
Classroom audio (opt-in)	Captured by mentor	New; consent from teacher and HT	E2	High → redact, delete audio ≤30 days DESIGN
Mentor notes / checklist	Captured by mentor	New	E2, E5	Medium
Practice-card library	SCERT/DIET, NCERT, DIKSHA [15]	Content exists; curation effort needed	E3, E4	Low
AI outputs + mentor edits	SAMVAAD	Generated	Evaluation, audit	Medium

The stack is our design choice DESIGN: Android app (offline SQLite, Opus audio) · Indic ASR — IndicConformer, 22 scheduled languages, MIT licence [16], fine-tuned on consented classroom audio · LLM: open-weight model self-hosted, or an API under a state MoU (decided at procurement) · Postgres + pgvector for retrieval · hosting on government-approved cloud · RBAC, encryption at rest and in transit, audit logs.

What happens when there is no internet

Night before · online
- Sync the day's plan
- Pre-generate practice cards + demo scripts
- Cache rubric in the language pack

At school · offline
- Record audio (opt-in)
- Checklist + voice notes
- On-device lookup of cached cards → feedback
- Nothing blocks the visit

Back in signal · sync
- Upload ~4.8 MB audio per 40 min (16 kbps)
- ASR + LLM run in cloud
- Report drafted; teacher voice note sent

63.5% of schools have internet, so 36.5% do not [3] DERIVED. States range from 18.6% (West Bengal) to over 90% (Kerala) [19b] (secondary analysis). The 4.8 MB figure is 16 kbps × 2,400 s DERIVED.

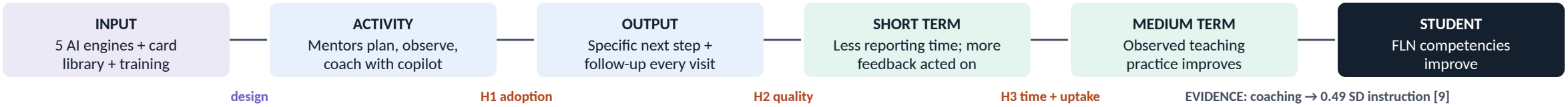
Privacy & responsible-AI controls DESIGN

- Teachers see their own record first. BEO and district see **aggregates only** (minimum group of 5 teachers)
- Coaching data is never used for appraisal, transfer or discipline. This is written into the state MoU
- Children's names redacted on device; no child-level AI outputs
- Compliance with the DPDP Act 2023 (consent, purpose limitation, retention) [20]
- Language-bias test: ASR and tagger error reported per language and dialect before each rollout

Sources: [2] Reach Alliance 2024 · [3] UDISE+ 2024–25 Table 1 p.29 · [10] Samagra 2024 · [15] PIB DIKSHA factsheet 2026 · [16] AI4Bharat IndicConformer model card · [19b] A.C. Mehta, educationforallinindia.com (UDISE+ analysis, secondary) · [20] Digital Personal Data Protection Act, 2023

A two-phase pilot that tests whether AI changes coaching before claiming it changes learning

Theory of change — each link is marked as evidence or hypothesis



Kraft et al. meta-analysis of 60 causal studies: coaching effects of 0.49 SD on instruction and 0.18 SD on achievement. Effects in larger programmes are "only a fraction" of those in small efficacy trials, and the studies are mostly from the US [9] FACT . Transferring these effects to Indian FLN mentoring is a HYPOTHESIS . **We claim no learning gain until it is measured.**

Pilot design ASSUMPTION on scale

Phase 0 — 8 weeks, 1 block, ~10 mentors. Tech validation on real classroom audio.

Phase 1 — one academic term (~6 months), 1 district. 60 mentors randomised within blocks (30 treatment / 30 business-as-usual). About 14 blocks at UP's average of ~4.4 ARPs per block (3,900 ÷ 880) [7] DERIVED . Caseloads unchanged.

Measurement: independent observers use a standard tool (e.g., World Bank Teach [17]) at baseline and endline on 12 teachers per mentor (720 teachers), plus mentor time-use logs and state FLN spot assessments (exploratory).

Power: MDES ≈ **0.30 SD** on observed practice (ICC 0.10, α 0.05, power 0.80). It falls to ≈0.25 SD with baseline covariates (R^2 0.3) DERIVED . Student learning is secondary and underpowered, and we say so.

KPI (LEVEL)	BASELINE	PILOT TARGET TARGET
Phase 0: tag agreement with DIET expert (AI)	—	Cohen's κ ≥ 0.6 on rubric items
Phase 0: drafts usable with minor edits (mentor)	—	≥ 70%
Visits ending in a written, specific next step (system)	Audit of current reports	≥ 80% of visits
Follow-up checked at next visit (mentor)	Audit	≥ 60%
Mentor time on reporting per visit (mentor)	Time-use log	−30% vs control
Teacher reports acting on feedback (teacher)	Survey	+15 pp vs control
Observed practice score (teacher)	Teach-type tool	≥ MDES (0.30 SD) vs control
Privacy incidents / teacher-score exposure	—	Zero (hard stop)

Go / no-go for scale: κ ≥ 0.6 and ≥70% usable drafts (Phase 0), then practice effect ≥ MDES *or* ≥30% time saving with a non-negative practice effect, ≥60% weekly active mentors, zero incidents. Targets are proposals, not findings.

Sources: [7] CSF 2026 (UP: ~3,900 ARPs, 880 blocks) · [9] Kraft, Blazar & Hogan, Review of Educational Research 88(4), 2018 · [17] World Bank, Teach Primary. Power calculation is a standard two-arm clustered MDES, z-sum 2.80, design effect 1+(m−1)ICC; the ICC is an assumption to be re-estimated at baseline.

Scale through the NIPUN structure that already exists, not a parallel system

PILOT
Phase 0 + 1, 60 mentors

BLOCK
All mentors in pilot district; BEO sees practice-gap view

DISTRICT
DIET owns card library & mentor training; DPMU dashboards take SAMVAAD aggregates [1 pp.29–30]

STATE
SCERT/ATF adopt the rubric and mentor mapping [1 p.23]; state-app integration; state-language ASR fine-tuning

MULTI-STATE
NCERT/NRG host an open rubric + practice-card standard [1 p.16]; content via DIKSHA [15]; each state runs its own deployment

Stays constant vs localised DESIGN

CONSTANT	LOCALISED PER STATE
Coaching loop and the 5 engines	Observation rubric items
Privacy rules; no teacher scoring	Language/dialect ASR packs
KPIs and evaluation protocol	Practice cards from SCERT textbooks
Open data schema	Integration with the state mentor app/MIS; cadre names (ABRC, BRP, CRP, ARP)

Cost model — variables, not invented prices

ASSUMPTION

$$\text{Cost/mentor/yr} = 12 \cdot V \cdot (A \cdot c_{asr} + K \cdot c_{llm}) + D + S + Tr + H$$

V = visits/month (UP assignment ≈30 [7]) · A = audio minutes/visit (≤40, observation block [6]) · K = tokens/visit · D = device share (existing phones first) · S = support · Tr = training · H = hosting share. Example load: 30 × 40 = **1,200 audio-min per mentor-month** DERIVED. Unit prices (c) are set at procurement; self-hosted open ASR [16] removes per-minute licence fees.

Why now

- NIPUN target: FLN by Grade 3 by **2026–27** [1] FACT
- NIPUN 2.0 agenda asks for structured mentor protocols [6]
- States already run mentor apps and speech-AI assessment [2] [10]
- Open Indic ASR covers all 22 scheduled languages [16]
- Coaching works but dilutes at scale [9]. AI is our proposed lever to hold quality at scale HYPOTHESIS

SAMVAAD makes a mentor's 20 minutes of feedback count: **one evidence-backed, followed-up next step for every teacher they visit**, inside the NIPUN system states already run.

Sources: [1] NIPUN Bharat Roles & Responsibilities, Foreword p.v, pp.16,23,29–30 · [2] Reach Alliance 2024 · [6] CSF 2026 · [7] CSF 2026 (UP) · [9] Kraft et al. 2018 · [10] Samagra 2024 · [15] PIB DIKSHA 2026 · [16] AI4Bharat IndicConformer

APPENDIX · NOT A CONTENT SLIDE

Source register (numbered references used in slides 1–10)

#	SOURCE	TIER / TYPE	URL / LOCATION · ACCESSED 2026-09-27
[1]	MoE, DoSE&L — *NIPUN Bharat Mission: Stakeholders: Roles and Responsibilities* (provided S1)	T1 official	Provided PDF; printed page numbers (PDF page = printed + 10)
[2]	Attavar, DeBacker, Kothuri, Lima, Ohyama, Mangla, Tudor — *Teacher Mentorship in India: Improving Pedagogy in Government Schools*, Reach Alliance, Jan 2024 (provided S2)	T2 university research (qualitative, Haryana)	Provided PDF; printed page numbers (PDF page = printed + 2)
[3]	MoE — UDISE+ 2024–25 Booklet, Table 1 National Highlights (pp.28–29)	T1 official data	dashboard.udiseplus.gov.in/…/UDISE+2024_25_Booklet_existing…pdf
[4]	ASER Centre — ASER 2024 National Findings (p.47), released Jan 2025	T1/T2 household survey (rural)	asercentre.org/wp-content/uploads/2022/12/ASER-2024-National-findings.pdf
[5]	Samagra Shiksha — New Programmatic & Financial Norms (monitoring indicators; BRC/CRC grants)	T1 official	samagrashiksha.in/…/download/AWP&B/norms.pdf
[6]	Central Square Foundation — “NIPUN Bharat Mission at Five: What Worked, What Next”, 23 Jun 2026	T2/T3 research org	centralsquarefoundation.org/articles/nipun-bharat-mission-at-five-…
[7]	CSF (Israni & Pandey) — “Navigating ARP Recruitment in Uttar Pradesh”, 22 Jan 2026	T2/T3 research org	centralsquarefoundation.org/articles/navigating-arp-recruitment-in-uttar-pradesh-…
[8]	Accountability Initiative (CPR), V. Davis — *Cluster Resource Centre Coordinators in Bihar* (n=4, 2014–16)	T2 research org (tiny sample)	accountabilityindia.in/sites/default/files/pdf_files/CRCC_Recommendations.pdf
[9]	Kraft, Blazar & Hogan (2018), Review of Educational Research 88(4):547–588	T2 peer-reviewed	doi.org/10.3102/0034654318759268
[10]	Samagra Governance — “Amrit Series #5: NIPUN Lakshya App”, 5 Jun 2024	T3 implementing partner	samagragovernance.in/blog/2024-06-05-amrit-series-5-nipun-lakshya-app/

APPENDIX · NOT A CONTENT SLIDE

Source register (continued) and research limitations

#	SOURCE	TIER / TYPE	URL / LOCATION · ACCESSED 2026-09-27
[11]	Piper et al. (2017), “Implementing large-scale instructional technology in Kenya”, IJEDICT 13(3):57–79	T2 peer-reviewed	files.eric.ed.gov/fulltext/EJ1166622.pdf
[12]	Microsoft Source Asia — “India’s schoolteachers are drafting better lesson plans faster…”, 8 Feb 2024	T4 company (used only for product description)	news.microsoft.com/source/asia/features/…copilot/
[13]	Demszky, Liu, Hill, Jurafsky, Piech (2024), EEPA — automated feedback RCT	T2 peer-reviewed	doi.org/10.3102/01623737231169270
[14]	Wang & Demszky (2023), “Is ChatGPT a Good Teacher Coach?”, BEA workshop, ACL	T2 peer-reviewed workshop	arxiv.org/abs/2306.03090
[15]	PIB — “DIKSHA: Powering India’s Digital Learning Ecosystem”, 28 Jun 2026	T1 official	pib.gov.in/FactsheetDetails.aspx?Id=150639
[16]	AI4Bharat — IndicConformer-600M-multilingual model card	T2 university lab	huggingface.co/ai4bharat/indic-conformer-600m-multilingual
[17]	World Bank — “Teach Primary: Helping countries…” brief	T2 multilateral	worldbank.org/en/topic/education/brief/teach-…
[18]	NITI Aayog Frontier Tech — “Data-Driven Classrooms: Gujarat’s Vidya Samiksha Kendra”, Jul 2023	T1 govt think-tank	frontiertech.niti.gov.in/story/data-driven-classrooms-…
[19] / [19b]	A.C. Mehta — “Introduction to BRC 2023” (provided); “63.5% of Indian Schools Now Online…” (UDISE+ analysis)	T3 secondary (ex-NIEPA)	educationforallinindia.com (both)
[20]	Digital Personal Data Protection Act, 2023 (Govt of India)	T1 law	Design constraint; legal review required

Research limitations

- Haryana evidence [2] is qualitative: 20 Zoom interviews (10 teachers, 5 mentors, 5 star mentors) in a few districts, with interviewees selected by the implementing partner, so selection bias is possible. It is state-specific, not India-wide.
- Bihar evidence [8] covers only 4 CRCCs in 2 districts (2014–16). It is illustrative, not representative.
- We found no nationwide dataset on mentor time use, visit quality or feedback follow-through. This is itself a gap the pilot will measure.
- The UP ratio [7] is an assignment norm, not observed practice. The Haryana ratio [2] is a calculated average.
- Coaching effect sizes [9] come mainly from US studies; automated-feedback evidence [13][14] comes from US and English contexts.
- The BRC essay [19] lists a state-wise “1,20,000 BRCs” count that is internally inconsistent (e.g., 45,875 for UP), so we excluded it.
- The explainer video link opened a sign-in-gated Google file titled “Challenge 4 Explainer”. **We could not access its content and did not use it.**